5.9.8

EE25BTECH11054 - S.Harsha Vardhan

Question 5.9.8 A father's age is three times the sum of the ages of his two children. After 5 years his age will be two times the sum of their ages. Find the present age of the father.

Solution: Let the present age of the father be F. Let the sum of the present ages of his two children be S.

$$F = 3S \tag{1}$$
$$F - 3S = 0 \tag{2}$$
$$F + 5 = 2(S + 10) \tag{3}$$
$$F + 5 = 2S + 20 \tag{4}$$
$$F - 2S = 15 \tag{5}$$

We now have a system of two linear equations with two variables, F and S. In matrix form $\mathbf{Ax} = \mathbf{b}$, this is:

$$\begin{pmatrix} 1 & -3 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} F \\ S \end{pmatrix} = \begin{pmatrix} 0 \\ 15 \end{pmatrix} \tag{6}$$

To solve this system, we use Gaussian elimination on the augmented matrix $\left(\mathbf{A} \mid \mathbf{b}\right)$.

$$\left(\begin{array}{cc|c} 1 & -3 & 0 \\ 1 & -2 & 15 \end{array}\right) \xrightarrow{R_2 \to R_2 - R_1} \left(\begin{array}{cc|c} 1 & -3 & 0 \\ 0 & 1 & 15 \end{array}\right) \xrightarrow{R_1 \to R_1 + 3R_2} \left(\begin{array}{cc|c} 1 & 0 & 45 \\ 0 & 1 & 15 \end{array}\right) \tag{7}$$

$$S = 15 \tag{9}$$

From the reduced row echelon form, we get the solution:

$$F = 45 \quad \text{and} \quad S = 15 \tag{11}$$

The question asks for the present age of the father. Thus, the present age of the father is 45 years.

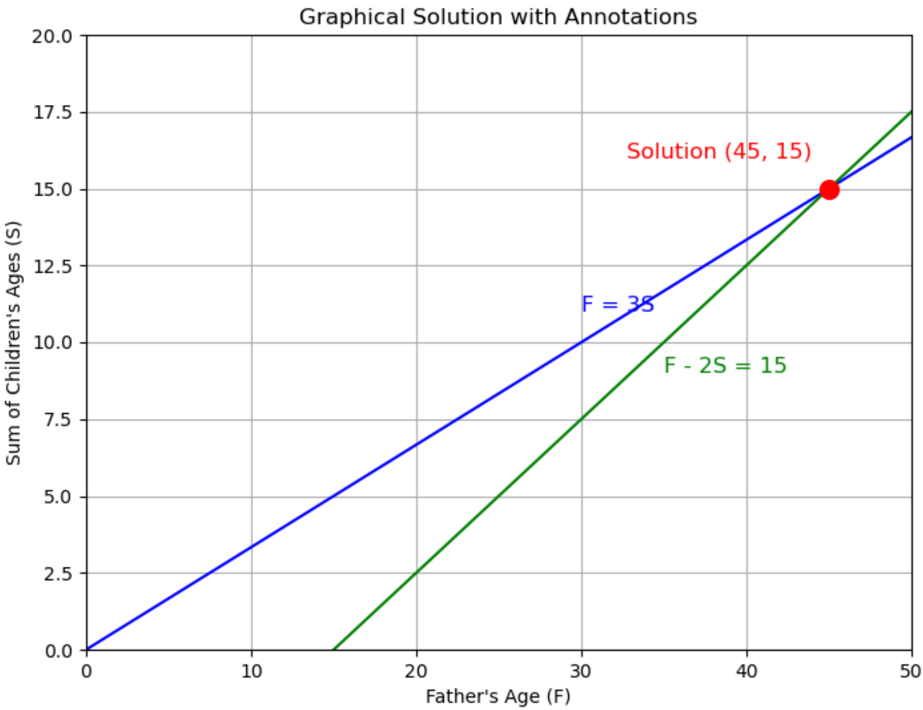
Graphical Solution with Annotations
Solution (45, 15)
F = 3S
F - 2S = 15
Sum of Children's Ages (S)
Father's Age (F)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0
10
20
30
40
50

Fig. 0.1